

Vadym Kurylenko

Curriculum Vitae

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Institute of Algebra and Geometry, OVGU Magdeburg

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Personal data

Date of birth October 1995

Nationality Ukrainian

Education

- 10/2020 – 12/2024 **PhD in Geometry and Mathematical Physics, SISSA, Trieste**
Thesis: Local Ehrhart Theory and Gale Duality.
Supervisor: Fernando RODRIGUEZ VILLEGAS.
- 10/2019 – 9/2020 **Doctoral studies, Universität Hamburg**
Supervisor Murad ALIM.
- 10/2017 – 9/2019 **Master's Degree in Mathematical physics, Universität Hamburg**
Thesis: Gauss-Manin Lie Algebra for Non-Compact Calabi-Yau Manifolds
Final mark: 1.1/1.
Supervisor: Murad ALIM.
- 9/2013 – 9/2017 **Bachelor's Degree in Physics, V.N. Karazin Kharkiv National University**
Thesis: The Regge-Wheeler Equation in a Quasi-Spherical Neutral Black Hole Background.
Final mark: 88/100, with honours.
Supervisor: Alexei NURMAGAMBETOV.

Positions

- 1/2025 – 1/2028 **Postdoc, Otto-von-Guericke Universität Magdeburg**
Mentor: Benjamin NILL.

Publications and preprints

Preprints

- 2025 **Examples of IDP lattice polytopes with non-log-concave h^* -vector**, with J. Hofscheier and B. Nill, <https://arxiv.org/abs/2505.18896>
- 2024 **Thin Simplices via Modular Arithmetic**, <https://arxiv.org/abs/2404.03975>

Publications

- 2022 **The algebra of derivations of quasi-modular forms from mirror symmetry**, with M. Alim and M. Vognin, <https://dx.doi.org/10.4310/PAMQ.2022.v18.n3.a6>

Talks

- 3/2025 **Combinatorial Synergies East, MPI, Leipzig**, (invited)
Local Ehrhart Theory and Its Synergies
- 2/2025 **Workshop on Polytopes, Universität Osnabrück**, (contributed)
Ehrhart Theory and Gale duality

- 7/2024 **Conference on Arithmetic Geometry and Applications**, *ICTP*, (contributed)
Ehrhart Theory and Gale Transforms
- 9/2023 **Workshop "Hodge and K-theory meet combinatorics"**, *EPFL*, (contributed)
Thin simplices via modular arithmetic.
- 5/2023 **Oberseminar in Magdeburg**, *OvGU Magdeburg*, (invited)
Thin simplices via modular arithmetic.
- 4/2023 **Graduate Student Meeting on Applied Algebra and Combinatorics**, *KTH, Stockholm*, (contributed)
Vanishing of local h^* -polynomial for simplices.
- 3/2023 **SISSA Junior GMP seminar**, *SISSA*
Hodge-Deligne Numbers of Hypersurfaces and Newton Polytopes.

Awards and Scholarships

- 10/2020 – 9/2024 **SISSA Fellowship**, 4-year fully funded scholarship
- 10/2017 – 9/2019 **DAAD Study Scholarship for Graduates of All Disciplines**, Scholarship for Master studies in Germany, c.100 recipients from Ukraine yearly.
- 2015 – 2017 **Scholarship of Verhovna Rada (Parliament) of Ukraine**, one nominee from the department and 500 scholarships to be awarded each year nationwide.

Visits

- 06/2023 **OvGU Magdeburg**, *Magdeburg*, Germany
Visitor as a guest of Prof. Benjamin Nill

Teaching

- Spring 2022 *ICTP*, Teaching assistant for Representation Theory
- 2014-2016 Coaching a school team for Regional Olympiads in Physics and Astronomy in Kharkiv

Scientific Interests

Ehrhart theory, hypergeometric functions, matroids, lattice width, mirror symmetry, enumerative geometry, modular forms.

Languages and Computer Skills

- Known Languages Russian (Native), Ukrainian (Native), English (Fluent), German (Level A2), Italian (Level A1).
- Computer Skills Sagemath, Python, Mathematica; basic familiarity with Polymake and Magma.